



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
FEBRUARY 2023**

Course : EC3247 (Cloud Architecture)

Time: 8.00 am – 11.00 am
(3 hours)

Lecturer: Muhammad Hafez Mohd Ariffin

Date: 23 May 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 4 printed pages.
(excluding the front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. List any **THREE (3)** common characteristics of Cloud Computing. (3 MARKS)
 - b. Explain the Hybrid Cloud with an appropriate diagram. (8 MARKS)
 - c. Assume that you are required to develop a user application for the Environment Monitoring System. This application will give user access to the real-time reading of environmental temperature, humidity, wind speed and air quality index. All this data will be stored in the cloud platform and can be accessed by user's hand phone. All the environmental sensors are part of Internet Of Thing (IOT) system based on Node MCU development kit. The programming language used is C# with wide of Node MCU standard library support.
 - i. Choose the best cloud delivery model to be used in the application. (1 MARK)
 - ii. Give the reasons for your selection of cloud delivery model. (2 MARKS)
 - iii. Clarify the **THREE (3)** important features that need to be considered to choose the best cloud service provider for the delivery models suggested in question 1 (c) (i). (6 MARKS)
2.
 - a. Describe the infrastructure- as- a- service (IAAS) (4 MARKS)
 - b. Suggest the dimension of scalability for the following scenarios: (4 MARKS)
 - i. An infrastructure-as-a-service (IaaS) instance in AliBaba Elastic Computing Services (ECS) have the auto-scaling function enable when the Virtual CPU (vCpu) can be increased in number when processing demand is increased.
 - ii. A Blynk platform-as-a-service (PaaS) services have the ability to upgrade the compiler once compiler speed is in high demand by IOT applications.
 - c. Describe the dimensions of scalability that you have suggested in b(i) and b(ii). (6 MARKS)

- d. Based on the following scenario, suggest the appropriate type of Cloud Delivery Model offered by the Cloud service provider: (6 MARKS)
- i. Google has application package containing Gmail, Google Meet and Google Calendar.
 - ii. Salesforce has CRM cloud service package for the clients to develop warehouse application.
 - iii. Amazon has service called Elastic Computing (EC2) for their clients to host their own webserver on the cloud.
 - iv. Arduino Cloud provides the cloud environment for IoT development for IoT systems.
 - v. Rackspace Cloud Servers offers a range of Linux-based pre-made images.
 - vi. MsTeam for Virtual Classroom by Microsoft
3. a. Explain any **THREE (3)** general benefits of Virtual Machines in cloud computing. (6 MARKS)
- b. You are about to select the suitable instances configuration in Alibaba ECS. There are many configurations of Virtual CPU (vCPU) cores provided in ECS to be selected according to the user preferences. Your computer programs have been developed in such way that 70% must be process in sequence (serial). The time taken to execute the whole program with a single CPU processor is 30 seconds.
- i. Using Amdahl's Law, calculate the expected speed up can be made to this program if you choose for 8 cores vCPU processors instances. (2 MARKS)
 - ii. Calculate the new time taken to execute this program. (2 MARKS)
 - iii. Suggest how to improve the speed up level. (2 MARKS)

C.

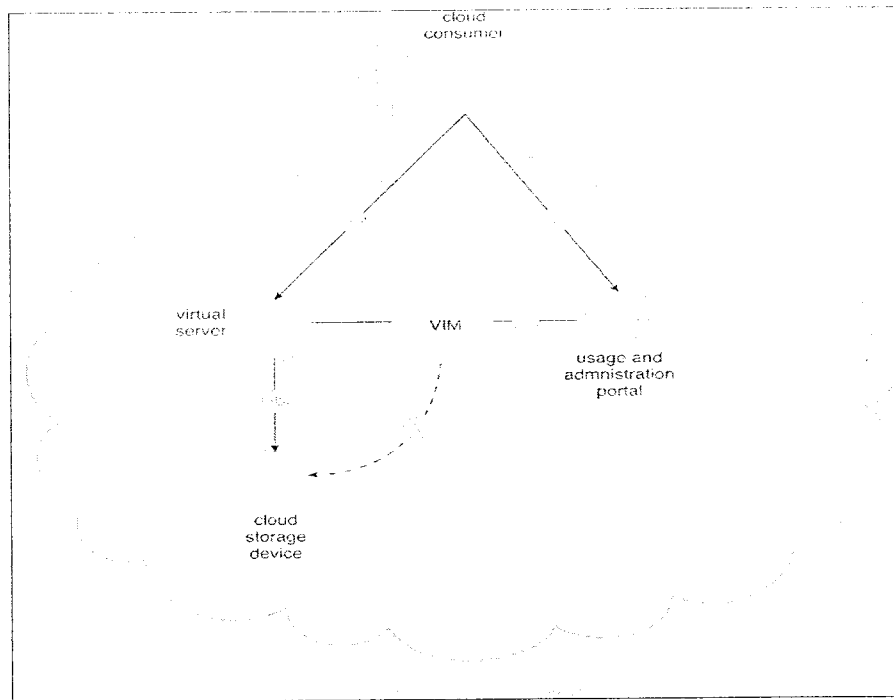


Figure 1

Figure 1 shows the way a cloud consumer gets an access to its cloud services inside virtual server. VIM is an important mechanism to enable Virtual environment in the cloud.

- i. Discuss the importance of VIM based on the given scenario. (4 MARKS)
- ii. Discuss any **TWO (2)** features of the VIM. (4 MARKS)

4. a. List the **THREE (3)** examples of Platform-as-a-services (PaaS) (3 MARKS)
- b. Briefly describe the following Challenges in platform-as-a- service (PaaS): (9 MARKS)
 - i. Information Security
 - ii. Data Lock-In
 - iii. Availability

- c. Amazon EC2 has a Web-based console user interfaces to access to instance mainly via SSH (Linux) and Remote Desktop (Windows). It also offers features like advanced reservation capacity (aka reserved instances) that guarantees availability for periods of 1 and 3 years. (4 MARKS)
Discuss the importance of the following features in Amazon EC2:
i. Variety of user interfaces and APIs.
ii. Advance Reservation of Capacity.
- d. Google Cloud Compute Engine (GCCE) services have divided its datacenter into region and zoning system. Discuss on why this feature is important for the infrastructure-as-a-services (IaaS). (4 MARKS)
5. a. Differentiate between the following Cloud Integration tools : (6 MARKS)
i. Traditional Cloud Integration tools
ii. Integration as a service
- b. DDoS attacks are major security risks in a cloud computing environment, where resources are shared by many users. A DDoS attack targets resources or services in an attempt to render them unavailable by flooding system resources with heavy amounts of unreal traffic. The objective of DDoS attacks is to consume resources, such as memory, CPU processing space, or network bandwidth, in an attempt to make them unreachable to end users by blocking network communication or denying access to services. Among other measures to be taken to overcome such issues is to have a disaster recovery and backup systems. (6 MARKS)
Discuss any **THREE (3)** good practices in order to have an efficient disaster recovery and backup system.
- c. CyberStream is a local startup company giving video streaming services to its subscribers. Initially this company operated with On-Premise webserver and video streaming server. The company's subscribers are on-rising and they forecast that their on premise facilities will not be able to cope with the demand in the next 6 months. They are planning to subscribe to Contents Delivery networks (CDN) services offered by the Ali Baba Cloud. Then the integration process with their on premise infrastructure will be arranged. (8 MARKS)
Discuss the **FOUR (4)** important steps for a successful cloud migration for the above scenario.

-END OF QUESTION PAPER-